

Experimental Evidence That Social Learning in Structured Information-Sharing Networks Corrects Anchoring Bias

Calvin Isch* and Damon Centola*

Abstract: The anchoring effect is a cognitive bias in which exposure to arbitrary initial values disproportionately influences estimates. While this effect has been widely documented, its role in social contexts remains unclear. Laboratory studies of isolated individuals often show stronger anchoring effects than naturalistic settings where people interact in social networks. We propose that social networks can amplify or attenuate anchoring depending on whether anchors are “helpful” or “harmful” for accuracy. In a preregistered experiment with 1600 participants across 40 groups, we compared individuals working independently to those embedded in peer-to-peer information-sharing networks. Networked groups improved overall belief accuracy, showing truth-seeking reductions in bias from harmful anchors, while also exhibiting truth-seeking increases in anchoring bias from helpful anchors. By contrast, isolated individuals showed no endogenous change in anchoring in either case. Our analyses identify a psychological mechanism that may underlie these effects: “confidence in others”, rather than self-confidence.

Key words: collective intelligence; anchoring effects; cognitive bias; networks; wisdom of the crowd; human-centered computing; collaborative and social computing

I Introduction

Anchoring effects are a robust class of psychological phenomena wherein decision-makers place disproportionate weight on arbitrary initial values when estimating quantities^[1–3]. Anchors have been found to influence consumer preferences^[4, 5], forecasts^[6, 7], legal decisions^[8], negotiation outcomes^[9, 10], and general numeric estimations^[11–13]. These effects have been identified in both experimental and field

settings and have been shown to persist even when individuals are offered large incentives to avoid them^[3, 14]. Recently, there has been increased interest in identifying strategies to reduce anchoring effects because of their potentially harmful impacts on increasing errors in human judgment^[3].

Here, we make two key contributions to this literature: (1) A focus on accuracy. Our experimental design moves beyond an analysis of the revision of beliefs toward or away from anchors and instead focuses on the change in accuracy associated with these belief revisions. (2) A focus on social learning. Our study is designed to evaluate how peer influences in information-sharing networks may either amplify or attenuate anchoring effects and their implications for belief accuracy. While anchors are often discussed in

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terms of their harmful effects on individual performance on estimation tasks (leading away from the true answer^[1–3, 6, 8]), there are nevertheless situations in which anchors may inadvertently help to improve the accuracy of individuals' judgements by pulling populations toward the true answer (Fig. 1).

As such, we classify anchors as “helpful” or “harmful” depending on whether they pull participants and groups toward or away from the true value. This classification strategy reflects a core feature of anchoring: Its effect depends on the relationship between an individual's prior belief, the correct value, and the anchor, rather than on any fixed property of the anchor itself. The same numerical anchor can provide informative evidence for one participant and misleading evidence for another. Our approach builds on this literature by making this dependency explicit: anchors are coded as helpful when they move estimates closer to the truth and harmful when they move them farther away. This approach aligns with models of anchoring effects as belief revision processes in which individuals partially update their priors toward new information when it appears credible^[15, 16], which also aligns with the psychological models that underly these effects^[17]. Our approach also extends recent work on anchoring in social settings^[18, 19] by embedding participants in information-sharing networks to assess how social learning within these networks shapes their belief revision process.

We build on prior work on network learning^[20–22], which has shown how information exchange in large social networks with uniform—i.e., egalitarian^[22, 23]—connectivity improves individual and collective

accuracy on estimation tasks^[22–25]. Studies of bias reduction have found that this process of collective learning in egalitarian networks can effectively reduce, and even eliminate political^[24, 26] and medical biases^[27, 28] in the evaluation of novel information. Here, we integrate this recent work on bias reduction in egalitarian information-exchange networks with the broad theoretical and empirical literature on anchoring effects. In keeping with the standard usage in the social learning literature, we define network learning as the process by which individuals iteratively update their beliefs based on information from connected peers. This framework follows the classic formulations of the DeGroot^[29] model of social learning, including subsequent extensions that analyze individual and collective belief revision processes in terms of social learning within peer-to-peer information-sharing networks^[22, 26, 30]. The mechanism that we identify underlying changes in collective beliefs could also be characterized as iterated peer influence (i.e., DeGroot belief updating), in which individuals repeatedly learn from neighbors' estimates, enabling “spillover” from neighborhood to neighborhood, which, across multiple rounds, indirectly influences neighborhoods beyond the immediate network ties. Given the small scale of our experimental networks, we do not claim evidence of cascades or heterogeneous structural effects.

Using a pre-registered experiment (See SI in the Electronic Supplementary Material (ESM) of the online version of this article, based on prior studies of network learning^[22, 23, 27]), we test the hypothesis that network learning dynamics within egalitarian information-exchange networks may have a truth-

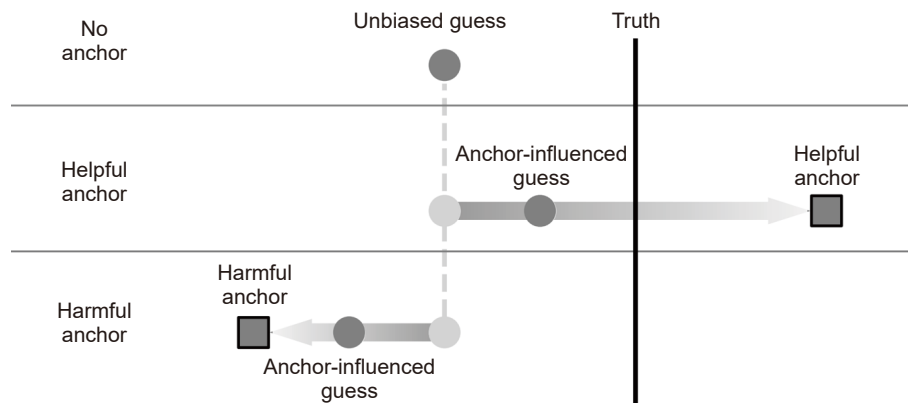


Fig. 1 Anchors can be helpful, pulling an unbiased estimate toward the true value, or harmful, pulling the estimate away from the truth.

seeking effect on populations subject to anchoring bias. For anchors that are in the opposite direction of truth relative to initial population estimates (see “harmful anchors” in Fig. 1), we hypothesize that information exchange within egalitarian networks will have a corrective effect, such that there will be significantly less anchoring bias in the population. However, when anchors are in the direction of truth relative to initial population estimates (see “helpful anchors” in Fig. 1), we expect information-exchange networks to demonstrate increased anchoring bias. In both cases, we expect that the effect of social influences within egalitarian networks will be to improve the accuracy of individuals’ estimates. Finally, we explore the network mechanisms by which individuals’ confidence may be used to explain network improvements in individual and collective reasoning in the face of anchoring bias. We find that self-reported confidence in one’s own answer is not correlated with error reduction, but self-reported “confidence in others” is.

2 Experimental Design

We recruited 1600 participants from Mechanical Turk to take part in a “Guessing Game” experiment, in which participants attempted to estimate the number of pennies contained in a large pile (see S2 “Experimental Design” in the ESM). Upon arriving to the study, subjects were randomized to either a social network condition or a control condition (see Fig. S1 in the ESM). They were also randomized to see either a high anchor stimulus or a low anchor stimulus (low = 118, high = 353, and truth = 246). The high and low anchor values were set at the 15th and 85th percentiles of responses from a separate randomly drawn population that completed the estimation task without exposure to any anchors (see Figs. S2–S4 in the ESM). With this 2x2 randomization, there were four conditions within each trial: Two network conditions (low and high anchor with 40 participants each), and two control conditions (low and high anchor also with 40 participants per group; see S3 “Statistical Analysis” in the ESM).

In both social network conditions, participants were placed into a decentralized, egalitarian network in which participants had uniform connectivity ($z = 4$ neighbors per person), which ensured that all participants had equal power (i.e., eigenvector centrality)

in the communication dynamics within the network. Network configurations were fixed such that connections did not change throughout the experiment. The same estimation task was used in every trial. Recruited subjects were only permitted to participate in this study once; thus, every trial contains a unique set of individuals. We conducted 10 trials in total ($N=160$ unique subjects per trial; $N=1600$ unique subjects in total). Across all conditions, subject compensation was contingent on the accuracy of their estimate in the final round.

Following previous studies of anchoring bias^[2, 3, 12], participants were first shown the image for the estimation task (a pile of pennies), and were then asked to assess whether their estimate was above or below an anchor value. This anchoring stage was followed by three rounds of responses. In round 1, all participants, regardless of experimental condition, provided an initial independent response to the estimation task (see Fig. S4 in the ESM). In the social network conditions in rounds 2 and 3, participants were shown the average of their neighbors’ responses from the previous round. They were then prompted to provide their response again and could either revise their response or enter the same answer used in the previous round. Participants viewed a peer signal equal to the mean of their four neighbors’ estimates. Consistent with prior work^[22, 26] we did not disclose the number of contributing peers. This choice helps prevent demand effects and strategic reweighting based on perceived sample size and supports generality across network manipulations without altering participants’ experience^[31].

The control condition followed a similar procedure; however, participants were not provided with any information about other participants’ responses. In rounds 2 and 3, participants in the control condition were given time to reflect on their answer and then prompted to provide their response. In both the control and network conditions, participants were given two minutes per round to reflect and revise their estimates. This time window was identical across conditions and across replicated trials to ensure that differences in anchoring and revision could not be attributed to unequal opportunities for deliberation.

In rounds 1 and 3, after participants completed the estimation task, they were also asked to report on

their “confidence in self” regarding the accuracy of their response in the current round. In addition, in round 3, participants in the social network condition were asked to report their “confidence in others”, indicating their confidence in the accuracy of the responses provided by their neighbors in the social network.

Importantly, our experiment supports causal inference only for factors that were randomly assigned: namely, assignment to the network versus control condition and exposure to high versus low anchors. For these comparisons (e.g., differences in changes in accuracy between networked and isolated participants), we indicate that these are causal results. By contrast, our analyses of harmful versus helpful anchors, and individual differences in confidence in self and others, rely on naturally occurring variation and are therefore presented as correlational results. These findings are to be interpreted as documenting robust patterns consistent with our theoretical framework, but not themselves as causal results.

Anchors can either move estimates toward the truth (helpful) or away from it (harmful in Fig. 1). We classify anchors by comparing the direction of the anchor relative to the initial guess R_1 and the truth τ . Let α be the anchor. If $(R_1 - \alpha)(R_1 - \tau) \geq 0$, the anchor is helpful ($f=1$); if $(R_1 - \alpha)(R_1 - \tau) < 0$, it is harmful ($f=0$). Formally:

$$f = \begin{cases} 1, & \text{if } (R_1 - \alpha)(R_1 - \tau) \geq 0; \\ 0, & \text{if } (R_1 - \alpha)(R_1 - \tau) < 0 \end{cases} \quad (1)$$

Because we do not observe participants’ unanchored estimates, this classification procedure is imperfect. In particular, we cannot detect cases where an anchor pushed a participant past the true value but increased accuracy; while in such a case anchors would move estimates closer to the truth, they would still be labeled as “harmful” because they lay on the opposite side of the true value compared to the initial unobserved estimate. We adopted this classification strategy to remain consistent with how anchors are typically operationalized in the literature^[3, 11]. Simulation analyses indicate that this labeling approach performs well for this task (see S4 “Classifying Anchors” in the ESM). Moreover, our proposed mechanism, a social-influence process that shifts responses toward the truth, yields opposite

predictions for helpful versus harmful anchors. To the extent that the labeling imperfectly captures participants’ responses, these cases tend to reduce differences between helpful and harmful anchors, pulling the two conditions toward one another. Consequently, our estimated effects are likely conservative. As a robustness check, we use this same procedure with the median network estimates and find the results do not differ qualitatively from those reported in the main text (see S3 “Statistical Analysis” in the ESM for crowd-level analysis).

Because participants in the network conditions were not statistically independent, our primary analyses were conducted at the group level. We measured both the changes in participants’ proximity to their respective anchors and the changes in their overall proximity to the true answer, both as a result of social influences. We first determined the absolute difference between every participant’s response at rounds 1 and 3 and their absolute distance to (1) their anchor and (2) the true answer (i.e., their absolute error). We then calculated the median of these values for each network and each control group, using these group-level measures as the observations for our statistical analysis. This conservative statistical approach reduces our statistical power to detect significant experimental effects. However, by accounting for the non-independence of individuals by restricting our analysis to group-level observations, our experimental approach (i.e., randomizing unique participants across all experimental conditions, and then replicating independent trials for all conditions) enables causal identification of the group-level effects of social influence on changes in anchoring bias. We provide additional analyses in the ESM demonstrating that our findings are supported (and strengthened) using traditional individual-level regression analyses with clustered standard errors (see S3 “Statistical Analysis” in the ESM).

3 Result

Subjects’ exposure to anchors during the initial anchoring stage of the study created significant anchoring effects. Participants’ round 1 responses in the low anchor condition were significantly different than participants’ responses in the high anchor conditions (low anchor median = 141; high anchor

median = 200; Mann-Whitney $p < 0.001$; unanchored median = 185). The anchoring index for these conditions, which is calculated as the difference between the anchored group medians divided by the difference between the anchors^[12], is 0.25 (see Fig. S5 for distributions by condition on rounds 1 and 3, Table S1 for distribution statistics by condition and round, and Table S2 for changes observed in each trial in the ESM).

Figure 2 shows the effects of social influence on changes in participants' distance to the anchor, for subjects exposed to helpful anchors ($N = 809$) and harmful anchors ($N = 635$). In both cases, over multiple rounds of revision, participants in the control condition exhibited no significant changes in their distance to their respective anchors.

By contrast, participants in the network conditions significantly adjusted their estimates in both cases. Participants exposed to harmful anchors adjusted their estimates to move significantly away from the anchor (increase in median distance away from anchor = 20.2, $p < 0.001$, Wilcoxon Signed-Rank, see Tables S3–S5 in the ESM), whereas participants exposed to helpful anchors adjusted their estimates to move significantly closer to the anchor (reduction in median distance to anchor = -14.1; $p < 0.001$; Wilcoxon Signed-Rank). Compared to participants in the control condition who were exposed to helpful anchors, participants in the network condition moved 13.6% closer to helpful anchors in networks ($p < 0.001$; Mann-Whitney U). Correspondingly, compared to control participants who were exposed to harmful anchors, participants in networks moved 28.7% farther from harmful anchors

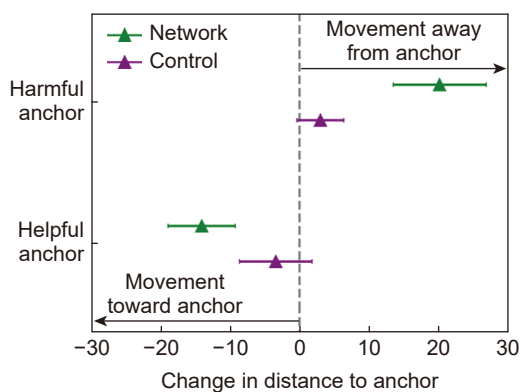


Fig. 2 Participants in the network conditions moved away from harmful anchors and closer to helpful anchors from rounds 1 to 3. Participants in the control conditions exhibited no significant change.

than the control participants did ($p = 0.004$; Mann-Whitney U; percentages reflect change in distance as a proportion of initial distance, see S3 “Statistical analysis” in the ESM for details on how percentage calculations were determined).

As we note in the introduction, what ultimately matters is not movement toward the anchor but how participants' estimates shift relative to the truth. Figure 3 shows that for all anchors, information-exchange in social networks significantly reduced errors in participants' estimates (see Table S2 in the ESM), a result that aligns with previous research^[22, 27, 28, 32, 33]. Relative to controls, participants in network conditions showed an overall reduction in error of 22% (change in error as a proportion of initial error, $p < 0.001$). The beneficial effects of information exchange in social networks were apparent for both participants who were exposed to helpful anchors (21% increase in error reduction relative to controls, $p < 0.001$), and for those exposed to harmful anchors (25% increase in error reduction relative to controls, $p = 0.005$; see Fig. S6 in the ESM).

The network mechanism that explains these truth-seeking collective dynamics, in which harmful anchoring bias is reduced, and helpful anchoring effects are amplified, is the disproportionate impact of accurate individuals in the process of belief revision within egalitarian social networks^[22, 26, 32]. As demonstrated in earlier studies of networked

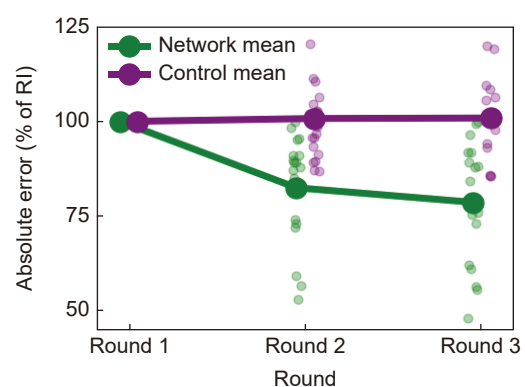


Fig. 3 Across both harmful anchors and helpful anchors, the absolute error of individuals' responses was substantially reduced in network conditions relative to control groups. The Y axis displays the error as a percentage of initial error. The absolute error is shown for each round and group, scaled proportionally to the Round 1 (R1) error. Lines and circles represent mean values across conditions; small points represent individual groups.

collective intelligence^[22, 27, 31], during the process of belief revision in egalitarian information-sharing networks there is an expected correlation between the accuracy of an individual's beliefs and the magnitude of their revisions, such that accurate individuals revise their responses less; this correlation between accuracy and revision magnitude is referred to as the “revision coefficient”^[22]. Within egalitarian social networks, a positive revision coefficient has been found to give greater de facto social influence to more accurate individuals, which is predicted to produce network-wide improvements in the accuracy of individual beliefs^[22, 26, 28, 31, 32]. We tested this prediction for participants in our study. The results show, as expected, that there is a significant positive revision coefficient among participants in the network condition ($p < 0.001$, Jonckheere-Terpstra Test, two-tailed) indicating that less accurate individuals made greater revisions to their responses while more accurate participants made smaller revisions (see Fig. S7 in the ESM). These analyses are robust to the exclusion of outliers (see S3 “Statistical Analysis” in the ESM for details on the partial correlation of the revision coefficient). As expected, networked

participants showed consistent network-wide improvements in accuracy, whereas isolated participants showed no change in average error (see Fig. S8 in the ESM for round-by-round belief change dynamics in networks and controls).

A secondary but important question concerns the psychological model of belief revision that underwrites these network learning dynamics^[22–24, 26–28, 31–33]. Past studies have explored whether a correlation between individual accuracy and self-reported confidence might explain why accurate individuals tend to revise their responses less^[34, 35]. However, evidence from several recent studies shows that self-reported confidence and individual accuracy are either uncorrelated^[36, 37], such that there is no clear relationship, or negatively correlated, such that individuals with greater self-reported confidence are less likely to be accurate^[23, 38, 39].

To address this puzzle, we first analyzed the relationship between individuals' self-reported confidence in their own answers and their accuracy on rounds 1 and 3, across all conditions. Consistent with previous studies, we found no significant correlation between participants' self-reported confidence and their accuracy in round 1 (see Fig. 4). Controlling for

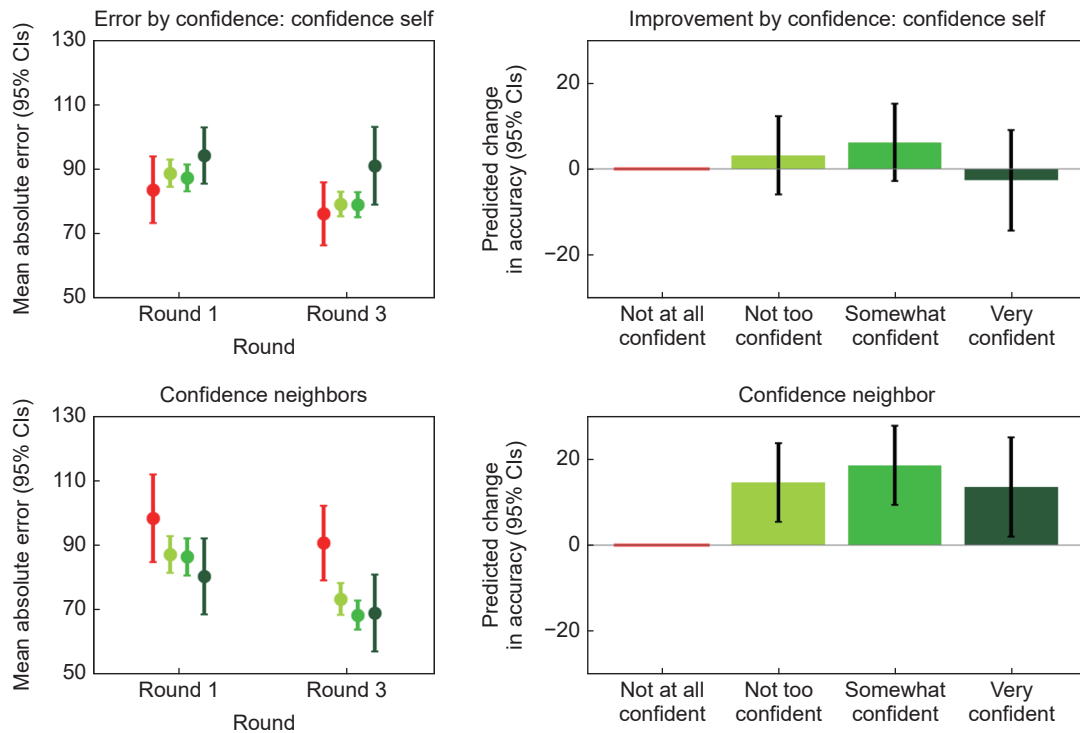


Fig. 4 Confidence in self (top row) was not associated with initial error or change in accuracy, whereas confidence in neighbors (bottom row) was associated with both lower initial error and greater improvements in accuracy (the difference in error from rounds 1 to 3).

initial accuracy, we also found that there was no significant relationship between changes in participants' accuracy (from rounds 1 to 3) and their self-reported confidence in their own responses (see S5 "Confidence Analysis" in the ESM). However, when we examined participants' self-reported confidence in others' responses, as reported in round 3, we found something different. There was a significant correlation ($p = 0.041$) between self-reported "confidence in others" and individual's initial (round 1) accuracy. We also found a significant relationship between self-reported "confidence in others" and increases in accuracy from rounds 1 to 3 ($p = 0.001$, F-Test, see Fig. 4). These findings indicate that participants who reported having more "confidence in others" were significantly more likely to be both initially accurate and to improve their accuracy during revisions (see Fig. S9 in the ESM for alternative visualization). We discuss these implications below.

4 Discussion

We found that learning in social networks can lead participants toward the true answer and reduce harmful anchoring effects. Unlike prior work, our approach focuses on the relationship between anchoring bias and the true answer, distinguishing between anchors that are harmful and those that are helpful. This conceptualization is consistent with work on bounded-rationality, in which scholars have suggested that the anchoring bias may reflect a rational use of cognitive resources providing useful information in many situations^[40–43]. Our study goes beyond these findings by showing that network learning can productively engage anchoring bias, dampening anchoring effects in some situations while amplifying them in others, in both cases moving the population closer to the true answer.

These findings help to explain why non-lab studies often find smaller anchoring effects than the more controlled lab experiments in which participants answer anchoring questions in isolation^[44–46]. In non-lab studies, individuals may be more likely to learn from their social network before forming their final estimate, resulting in accurate estimates that are farther from the harmful anchors; i.e., a reduced anchoring effect. Our experimental approach may therefore offer improved ecological validity for

identifying how anchoring effects actually influence individuals' judgements in uncontrolled environments.

As with any experiment, the features of our experimental design that enabled causal identification also limited the range of behaviors that we could test. In particular, we utilized a single estimation task to allow for a clear comparison of results across replicated experimental trials. Past experimental work on network learning found that using multiple question formats produced results that were comparable to single-questions^[22]; nevertheless future studies may explore whether changes in question topics might offer productive extensions of our findings. Additionally, the anchoring index for our stimulus was relatively small (0.25) such that the gap between the low and high anchored conditions could be substantially higher^[12]. We utilized this stimulus because it enabled us to compare our findings to previous studies using this knowledge question, for which baseline network learning dynamics for the distribution of responses without anchoring had already been established^[22]. Our results may thus offer a conservative estimate of the corrective effects of network learning dynamics. We anticipate that studies exhibiting a larger anchoring index (i.e., a greater difference between high and low anchors) will offer greater opportunities for improvement via social network interactions and will therefore exhibit stronger improvements than those found here.

Consistent with prior work, we found no relationship between self-reported confidence in self and individual accuracy^[23, 36–38]. By contrast, we found that "confidence in others" was significantly correlated with both higher initial accuracy and larger accuracy gains during revision. This finding is interesting because, as shown above, individuals with greater initial accuracy tend to revise their answers less. Thus, our results suggest that there may be two complementary mechanisms underlying our findings on subjects' reports of confidence in others, which may independently affect different sub-populations. First, for individuals who are initially accurate, the average of neighborhood responses (which report a small scale version of the wisdom of the crowd^[20, 47]) is likely to be more accurate than most individuals, and thus more similar to the accurate individual's response. Thus, an initially accurate individual may report high "confidence in others" as a result of the close

proximity between their initial response and the average responses of their peers. In which case, although an initially accurate individual may not revise their answer in response to social information, they may nevertheless express confidence in their peers. (A corresponding observation by Asch (1951) shows that when highly accurate people do not go along with majority view held by their peers, they will continue to express high regard for their peers' responses, such as high "confidence in others"[48].)

With regard to the second group of people— i.e., individuals who were initially less accurate, and thus revised their responses by a large magnitude between rounds 1 and 3—their pattern of responses suggests a simple mechanism for their self-reported "confidence in others". Namely, self-reported "confidence in others" functions as metacognitive description of how much weight these individuals place on their own opinion versus the opinions of others[22–24, 26–28, 31–33]. These initially-inaccurate individuals expressed higher confidence in others because it reflects their reliance on the social signals from their peers to improve their estimate — this reliance is an ecologically rational heuristic given that aggregating partially independent estimates tends to move collective judgments toward the truth[37, 39, 49]. Our results extend this principle from dyads to networked settings, showing that limited local exchange can amplify these benefits.

Both of these mechanisms have scope conditions. They depend on expected gains from aggregated peer learning, and the "wisdom of the crowd" improvements that arise from informational diversity. Homophily may be an important scope condition. In highly homophilous networks, information aggregation may not be as productive if individuals all share the same information. Thus while homophily may increase "confidence in others" it may be correlated with less improvement in network learning. Contrasting this observation, recent work has challenged the assumption that homophily limits collective intelligence, showing instead that network learning in egalitarian structures can still be effective for increasing collective intelligence and reducing group bias, even within homogenous "echo chambers"[23]. In light of these studies, there are many interesting avenues to explore regarding the relationship between "confidence in others" and network learning. Future work should

vary network structure, prior diversity, and confidence calibration to evaluate various conditions under which "confidence in others" improves versus undermines collective accuracy.

These findings open up several important new avenues of research. Past studies have shown that individual accuracy changes across questions, as does self-reported confidence. Does "confidence in others" change across questions, too? Past studies have also shown that the relationship between self-confidence and accuracy changes with the difficulty of the task[39, 41]. Our results suggest that this can be explained by the relationship between consensus view and the true answer. Does self-reported "confidence in others" also change in the predicted way with the difficulty of the estimation task? These questions offer interesting new directions for future research. Additionally, although our analyses focus on collective-level accuracy, anchors and peer influence may have heterogeneous effects across individuals. A limitation of this study is that we did not collect demographic information that might be useful for such analyses. Prior work suggests that anchoring effects depend on individuals' knowledge levels[50] and that conformity varies across both age and culture[33, 51, 52]. Moreover, as our study used an online survey sample, participants may reflect WEIRD populations that limit generalizability across cultures[53]. We suggest that future work might benefit from testing how individual and cultural factors might interact with network learning processes to influence collective outcomes.

Although our evidence comes from a laboratory setting, the results speak to anchoring in real decision environments. First, by distinguishing helpful from harmful anchors, the findings align with an "ecological rationality" perspective, which treats anchoring as context-dependent rather than uniformly detrimental[43]. Second, because many consequential judgments are made in teams or online groups, our results suggest that egalitarian information-sharing networks can mitigate the adverse effects of harmful anchors that are observed among isolated individuals. Finally, the findings have design implications: collaborative platforms may perform best when they support decentralized, egalitarian exchange rather than amplifying hierarchical influences that may promote anchoring and other biases[32].

Finally, our findings offer a new perspective on dual-process accounts of cognition, which suggest that people can overcome biased judgments (i.e., “fast-thinking” or “System 1” judgments) through intentionally engaging more deliberate cognitive processes (i.e., “slow thinking”, or “System 2” judgments)^[54]. Our design explores this idea by implementing control conditions in which participants were explicitly given the opportunity to revise their estimate after isolated reflection. We found that despite this opportunity for independent deliberation, participants in the control conditions did not show significant improvements in their responses, nor any tendency to overcome their anchoring bias. By contrast, individuals in information-exchange networks, who had reduced capacity for independent reflective judgment, nevertheless enjoyed significant reductions in overall error, and were able to reduce harmful anchoring effects. These results suggest that in addition to “System 1” and “System 2”, perhaps there might be additional cognitive processes that are specifically engaged within socially embedded choice settings. Today, it is also an open question to what extent social network learning dynamics, as compared to independent reflection, directly engage System 1 versus System 2 reasoning processes^[28]. We expect that an interesting direction for future studies will be the identification of cognitive mechanisms through which social learning dynamics can lead to bias reduction in populations that are explicitly subject to cognitive biases, such as anchoring effects.

Data and Code Availability

The code for this study was written in Python and R, and is deposited on GitHub, available at <https://github.com/CalvinIsch/CI-Cognitive-Bias>, along with the data required for full reproduction.

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Conflict of Interest

The authors declare no conflict of interest.

Electronic Supplementary Material

Supplementary materials are available in the online version of this article at <https://doi.org/10.23919/JSC.2026.0005>.

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